

Dragon Ball: The Breakers API services

Bandai Namco backend

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Abstract

This paper analyzes the application-layer communication used by *Dragon Ball: The Breakers* between the Steam client and Bandai Namco's Cosmos Channel backend services. Based on captured HTTP/2 traffic, the analysis identifies HTTPS transport, MessagePack-encoded application data, and two backend services: a common User/System API and a game-specific API.

The observed requests share a common structure containing the title code, user identifier, session, platform, and, for game-specific requests, a version field. Responses contain result metadata and an evolving session value. Several endpoints were reconstructed, including authentication, country lookup, tracking-number retrieval, user-information processing, maintenance information, and adjustment-data retrieval.

The analysis establishes the observed protocol structure while identifying several endpoint-specific fields and the large adjustment-data payload as areas requiring further investigation.

Contents

1	Introduction	4
2	Structures	5
2.1	HTTP	5
2.1.1	Request	5
2.1.2	Response	5
2.1.3	Exception	6
2.2	MessagePack	7
2.2.1	Example	7
2.2.2	Request	8
2.2.3	Response	9
3	API	10
3.1	Overview	10
3.2	Session	11
4	API Calls	12
4.1	System and Account	13
4.1.1	sys/agree_kpi	14
4.1.1.1	Request	14
4.1.1.2	Response	14
4.1.2	sys/get_env_v3	15
4.1.2.1	Request	15
4.1.2.2	Response	16
4.1.3	user/auth	17
4.1.3.1	Request	17
4.1.3.2	Response	18
4.1.4	user/get_country	19
4.1.4.1	Request	19
4.1.4.2	Response	19
4.1.5	user/get_tracking_num	20
4.1.5.1	Request	20
4.1.5.2	Response	20
4.1.6	user/create_user_info	21
4.1.6.1	Request	21
4.1.6.2	Response	22
4.2	Bandai Namco ID	23
4.3	Operational and Administrative Functions	24
4.3.1	close/get_close_info	25
4.3.1.1	Request	25
4.3.1.2	Response	26
4.3.2	adjustment_data_manage/read	28
4.3.2.1	Request	28
4.3.2.2	Response	29
4.3.2.3	Payload	30
4.4	Friends and Social	31
4.5	Blocking	32
4.6	Battle	33
4.6.1	battle/get_stun_server_info	34

4.6.1.1	Request	34
4.6.1.2	Response	35
4.7	Events	36
4.8	Friend Codes	37
4.9	Lootbox	38
4.10	Leaderboard	39
4.11	Messages	40
4.12	News	41
4.13	Character and Cosmetic Data	42
4.14	Transball and Skills	43
4.15	Rival and Patroller Systems	44
4.16	Player and Economy	45
4.17	Season	46
4.18	Shop	47
4.19	Common Purchase	48
4.20	Online Battle and Connection Data	49
4.21	Miscellaneous Battle Configuration	50
5	End-to-End Request Sequence	51
6	Conclusions	52
7	Reverse engineering	53
7.1	Executable	53
7.1.1	API	53
7.1.1.1	Class Architecture	53
7.1.1.2	Class list	54

1 Introduction

The captured traffic represents application communication between a client using the Steam HTTP client implementation and servers associated with `cosmos.channel.or.jp`.

The communication stack is:

```
TLS
|
+-- HTTP/2
    |
    +-- HEADERS
        |   +-- :method
        |   +-- :path
        |   +-- :scheme
        |   +-- :authority
        |   +-- user-agent
        |   +-- content-type
        |   +-- accept
        |   +-- accept-encoding
        |   +-- accept-charset
        |   +-- content-length
        |
    +-- DATA
        |
        +-- MessagePack
```

The packet captures were inspected at the point where Wireshark had already decoded the TLS traffic and HTTP/2 headers. Therefore, this paper focuses on application-level information rather than Ethernet, TCP, TLS record framing, or HPACK implementation details.

2 Structures

2.1 HTTP

All observed application requests are HTTP/2 POST requests.

2.1.1 Request

The common request headers are:

```
:method: POST
:path: <endpoint>
:scheme: https
:authority: <server>
user-agent: Valve/Steam HTTP Client 1.0 (1276760)
content-type: application/x-www-form-urlencoded
accept: text/html,*/*;q=0.9
accept-encoding: gzip,identity,*,q=0
accept-charset: ISO-8859-1,utf-8,*,q=0.7
content-length: <request length>
```

2.1.2 Response

The response headers generally contain:

```
:status: 200
server: nginx
date: <server date>
content-type: application/x-messagepack; charset=utf-8
content-length: <response length>
access-control-allow-origin: *
via: 1.1 google
alt-svc: h3=":443"; ma=2592000
```

The use of:

```
content-type: application/x-messagepack; charset=utf-8
```

on the response explicitly identifies MessagePack as the application serialization format.

2.1.3 Exception

Although the HTTP request header uses:

```
content-type: application/x-www-form-urlencoded
```

the captured request DATA is MessagePack encoded.

Conversely, the server explicitly reports:

```
content-type: application/x-messagepack; charset=utf-8
```

in responses.

Consequently, the effective application protocol is:

```
HTTP/2 POST
|
+-- request headers
|
+-- MessagePack DATA
|
v
Server
|
+-- HTTP/2 response headers
|
+-- MessagePack DATA
```

The request-side content type therefore appears to describe the transport interface expected by the original client rather than the actual encoding of the captured application body.

2.2 MessagePack

2.2.1 Example

MessagePack data is been encode to binary data and decoded to json data.

This is how a HTTP packet would look like with MessagePack data inside:

0000	00 00 23 01 04 00 00 00 05 83 04 94 60 00 00 00	..#.....'...
0010	06 07 59 98 b5 05 b1 89 8a 98 88 7b 6a 4d 9e bf	..Y.....{jM..
0020	87 c3 c2 c1 c0 bf be 0f 0d 02 37 36 00 00 4c 00	76..L.
0030	01 00 00 00 05 92 84 a7 74 69 74 6c 65 43 64 a6	titleCd.
0040	30 32 35 33 34 38 a6 75 73 65 72 49 64 b2 36 38	025348.userId.68
0050	35 31 32 39 32 36 30 35 32 32 31 39 32 35 34 39	5129260522192549
0060	a7 73 65 73 73 69 6f 6e ad 36 61 31 30 61 64 62	.session.6a10adb
0070	64 36 35 61 65 63 a8 70 6c 61 74 66 6f 72 6d 03	d65aec.platform.
0080	90	.

The binary MessagePack data starts at 0x92.

You can copy this data as a hex stream in wireshark as example:

```
9284a77469746c654364a6303235333438a6757365724964b236383531323932
3630353232313932353439a773657373696f6ead366131306164626436356165
63a8706c6174666f726d0390
```

And you can decode it to:

```
[
  {
    "titleCd": "025348",
    "userId": "685129260522192549",
    "session": "6a10adbd65aec",
    "platform": 3
  },
  []
]
```

Info

You can use this online tool to decode MessagePack data:
<https://ref45638.github.io/msgpack-converter/>

2.2.2 Request

A recurring request object was observed throughout the API.

The general form is:

```
[
  {
    "titleCd": "...",
    "userId": "...",
    "session": "...",
    "platform": 3,
    "version": "09.01"
  },
  <endpoint-specific arguments>
]
```

Not every request contains the **version** member. The earlier /000000/ system and user APIs use a smaller common object.

For example:

```
[
  {
    "titleCd": "025348",
    "userId": "685129260522192549",
    "session": "6a10adb...",
    "platform": 3
  },
  []
]
```

For the game-specific /025348/ APIs, the observed common object is:

```
[
  {
    "titleCd": "025348",
    "userId": "685129260522192549",
    "session": "6a10ad...",
    "platform": 3,
    "version": "09.01"
  },
  []
]
```

The second element of the outer array contains endpoint-specific arguments. Several endpoints use an empty array, indicating that no additional endpoint arguments were necessary in those requests.

2.2.3 Response

The observed responses share a common outer structure:

```
[
  {
    "result": 0,
    "date": "...",
    ...
  },
  <endpoint-specific data>
]
```

3 API

3.1 Overview

Two API servers are been used for the game.

Service name	id	Domain name
User/System API	000000	cosmos.channel.or.jp
Game API	025348	dbtb-prd.cosmos.channel.or.jp

The User System API services is for mostly basical API functionality that all Bandai Namco games are using. And the Game API services is been used for specifcly "Dragon ball - The breakers"

3.2 Session

An important feature of the observed protocol is that the session value is present in both requests and responses.

The captured sequence demonstrates that the server can return a new session value which is subsequently used by later requests.

A simplified sequence is:

```
AUTH
  request session = ""
  response session = 6a10adbd65aec

GET COUNTRY
  request session = 6a10adbd65aec
  response session = 6a10adbddb540

GET TRACKING NUMBER
  request session = 6a10adbddb540
  response session = 6a10adbe3e419

GET CLOSE INFO
  request session = 6a10adbe3e419
  response session = 6a10adbecae8c
```

The later game-service requests continue this pattern.

This strongly suggests that the session field is not merely an identifier provided once during login. It behaves as evolving application state.

4 API Calls

The observed API calls can be summarized as follows. The following table provides an overview of the identified server API endpoints.

4.1 System and Account

Endpoint	Function description
sys/agree_kpi	KPI agreement
sys/check_nname	Name restriction check
sys/get_env_v3	Environment/configuration retrieval
sys/kpi	KPI information
user/auth	User authentication and session establishment
user/get_country	Country lookup
user/get_tracking_num	Tracking-number retrieval
user/create_user_info	User information creation/update
user/update_manner_point	Manner-point update
user/get_ban_status	Account ban-status retrieval
user/update_avatar	Avatar update

4.1.1 sys/agree_kpi

4.1.1.1 Request

Observed request:

```
POST /000000/api/sys/agree_kpi HTTP/2
:authority: cosmos.channel.or.jp
content-type: application/x-www-form-urlencoded
content-length: 64
```

The request contained the common user fields and no additional fields.

```
[
  {
    "titleCd": "025348",
    "userId": "",
    "session": "",
    "platform": 3
  },
  []
]
```

4.1.1.2 Response

The response was:

```
HTTP/2 200 OK

content-type: application/x-messagepack; charset=utf-8
content-length: 80
```

The response contained the common user fields and with additional field.

```
[
  {
    "result": 0,
    "date": "2026/05/22 19:26:34",
    "session": ""
  },
  [
    0
  ]
]
```

The endpoint-specific information includes:

```
unknown      = 0
```

Warning

The first integer value, 0, is currently of unknown purpose. Further research is required to determine its intended use.

4.1.2 sys/get_env_v3

4.1.2.1 Request

Observed request:

```
POST /000000/api/sys/get_env_v3 HTTP/2
:authority: cosmos.channel.or.jp
content-type: application/x-www-form-urlencoded
content-length: 547
```

The request contained the common user fields and with additional fields.

```
[
  {
    "titleCd": "025348",
    "userId": "",
    "session": "",
    "platform": 3
  },
  [
    256,
    "",
    1,
    ""
  ]
]
```

The endpoint-specific information includes:

```
unknown = 256
unknown = ""
unknown = 1
unknown = ""
```

Warning

All the values are currently of unknown purpose. Further research is required to determine its intended use.

4.1.2.2 Response

The response was:

```
HTTP/2 200 OK

content-type: application/x-messagepack; charset=utf-8
content-length: 80
```

The response contained the common user fields and with additional fields.

```
[
  {
    "result": 0,
    "date": "2026/05/22 19:25:49",
    "session": ""
  },
  [
    0,
    "https://cosmos.channel.or.jp/",
    "https://dbtb-prd.cosmos.channel.or.jp/"
  ]
]
```

The endpoint-specific information includes:

```
unknown          = 0
api_user_system = "https://cosmos.channel.or.jp/"
api_game         = "https://dbtb-prd.cosmos.channel.or.jp/"
```

Warning

The first integer value, 0, is currently of unknown purpose. Further research is required to determine its intended use.

4.1.3 user/auth

4.1.3.1 Request

Observed request:

```
POST /000000/api/user/auth HTTP/2
:authority: cosmos.channel.or.jp
content-type: application/x-www-form-urlencoded
content-length: 547
```

The request contained the common user fields with additional fields.

```
[
  {
    "titleCd": "025348",
    "userId": "",
    "session": "",
    "platform": 3
  },
  [
    "76561198699994862",
    "14000000167B1B601A3B52DEEE5E172C
      01001001A0AD106A1800000001000000
      02000000AE7C4377DC43E19E8E120300
      02000000B80000003800000004000000
      EE5E172C01001001587B1300E7EF2BD9
      CA01A8C002000000A1AD106A215D2C6A
      0100706A0B0001006C1E1B0000000000
      8B5C28653F02A55FA15FF261944218EC
      1E067826FF6E844F9110A783FCD9584C
      27466EEF8ABD94BA34B29A4DA7A8690D
      6E1AC24BD339942340FE86DD31B2E89E
      21EB1F099E57097ACAE47D3800E8C825
      787327D09E75DA6B132D7C07A211AC88
      4A399BC751D9CE7B6DB494CC6FBA2980
      27743BEA0CB679254F902F5E615CCCC6",
    0
  ]
]
```

The endpoint-specific information includes:

```
steam_id    = "76561198699994862"
steam_auth  = "14000000167B1B601A3B52DEEE5E172C..."
unknown     = 0
```

Warning

The Steam authentication data is unknown how this is been generated and how the API server can check. This need further research how this data can be validated.

Warning

The third integer value, 0, is currently of unknown purpose. Further research is required to determine its intended use.

4.1.3.2 Response

The response was:

```
HTTP/2 200 OK

content-type: application/x-messagepack; charset=utf-8
content-length: 80
```

The response contained the common user fields and with additional fields.

```
[
  {
    "result": 0,
    "date": "2026/05/22 19:25:49",
    "session": "6a10adbd65aec"
  },
  [
    0,
    [
      "685129260522192549",
      1
    ]
  ]
]
```

The endpoint-specific information includes:

```
unknown = 0
info = [
  user_id   = "685129260522192549"
  unknown   = 1
]
```

The client will set the UserId to its session in future requests.

Warning

The unknown values are currently of unknown purpose. Further research is required to determine its intended use.

4.1.4 user/get_country

4.1.4.1 Request

```
POST /000000/api/user/get_country HTTP/2
:authority: cosmos.channel.or.jp
content-type: application/x-www-form-urlencoded
content-length: 76
```

The request contained the common user fields and no additional fields.

```
[
  {
    "titleCd": "025348",
    "userId": "685129260522192549",
    "session": "6a10adbd65aec",
    "platform": 3
  },
  []
]
```

4.1.4.2 Response

```
HTTP/2 200 OK

content-type: application/x-messagepack; charset=utf-8
content-length: 62
```

The response contained the common user fields with additional fields.

```
[
  {
    "result": 0,
    "date": "2026/05/22 19:25:49",
    "session": "6a10adbddb540"
  },
  [
    0,
    ["GB"]
  ]
]
```

The endpoint-specific information includes:

```
unknown = 0
country = "GB"
```

Warning

The first integer value, 0, is currently of unknown purpose. Further research is required to determine its intended use.

4.1.5 user/get_tracking_num

4.1.5.1 Request

```
POST /000000/api/user/get_tracking_num HTTP/2
:authority: cosmos.channel.or.jp
content-type: application/x-www-form-urlencoded
content-length: 76
```

The request contained the common user fields and no additional fields.

```
[
  {
    "titleCd": "025348",
    "userId": "685129260522192549",
    "session": "6a10adbddb540",
    "platform": 3
  },
  []
]
```

4.1.5.2 Response

```
HTTP/2 200 OK

content-type: application/x-messagepack; charset=utf-8
content-length: 76
```

The response contained the common user fields with additional fields.

```
[
  {
    "result": 0,
    "date": "2026/05/22 19:25:50",
    "session": "6a10adbe3e419"
  },
  [
    0,
    "LXDA-45J98PDC7NL"
  ]
]
```

The endpoint-specific information includes:

```
unknown = 0
unknown = "LXDA-45J98PDC7NL"
```

Warning

All the values are currently of unknown purpose. Further research is required to determine its intended use.

4.1.6 user/create_user_info

4.1.6.1 Request

```
POST /025348/api/user/create_user_info HTTP/2
:authority: dbtb-prd.cosmos.channel.or.jp
content-type: application/x-www-form-urlencoded
content-length: 114
```

The request contained the common user fields with additional fields.

```
[
  {
    "titleCd": "025348",
    "userId": "685129260522192549",
    "session": "6a10adbecae8c",
    "platform": 3,
    "version": "09.01"
  },
  [
    "GB",
    "en",
    "76561198699994862"
  ]
]
```

The endpoint-specific information includes:

```
country = "GB"
language = "en"
steam_id = "76561198699994862"
```

4.1.6.2 Response

The server returned:

```
HTTP/2 200 OK

content-type: application/x-messagepack; charset=utf-8
content-length: 84
```

The response contained the common user fields with additional fields.

```
[
  {
    "result": 0,
    "date": "2026/05/22 19:25:52",
    "version": "09.01",
    "flag": "0",
    "session": "6a10adbf368d5"
  },
  [
    0,
    "GB",
    1
  ]
]
```

The endpoint-specific information includes:

```
unknown = 0
country = "GB"
unknown = 1
```

Warning

The first and last value are currently of unknown purpose. Further research is required to determine its intended use.

4.2 Bandai Namco ID

Endpoint	Function description
bnid_client/check_status	BNID link-status check
bnid_client/entry	BNID linking/entry operation
bnid_client/erase	BNID link removal
bnid_reward/grant_reward	BNID reward grant

4.3 Operational and Administrative Functions

Endpoint	Function description
regularly_run/regularly_maintenance_check	Maintenance-status check
regularly_run/get_emergency_message	Emergency message retrieval
report/report	Unfair-play/report submission
close/get_close_info	Close/maintenance information
adjustment_data_manage/get_version	Online battle data version retrieval
adjustment_data_manage/read	Online battle adjustment-data retrieval

4.3.1 close/get_close_info

4.3.1.1 Request

The request was sent to a different host:

```
POST /025348/api/close/get_close_info HTTP/2
:authority: dbtb-prd.cosmos.channel.or.jp
content-type: application/x-www-form-urlencoded
content-length: 93
```

The request contained the common user fields with additional fields.

```
[
  {
    "titleCd": "025348",
    "userId": "685129260522192549",
    "session": "6a10adbe3e419",
    "platform": 3,
    "version": "09.01"
  },
  [
    "en"
  ]
]
```

The endpoint-specific information includes:

```
language = "en"
```

4.3.1.2 Response

```
HTTP/2 200 OK
```

```
content-type: application/x-messagepack; charset=utf-8  
content-length: 177
```

The response contained the common user fields with additional fields.

```
[  
  {  
    "result": 0,  
    "date": "2026/05/22 19:25:50",  
    "version": "09.01",  
    "flag": "0",  
    "session": "6a10adbecae8c"  
  },  
  [  
    0,  
    "",  
    [  
      [  
        1,  
        "2100-01-01 07:00:00"  
      ],  
      [  
        2,  
        "2100-01-01 07:15:00"  
      ],  
      [  
        3,  
        "2100-01-01 07:25:00"  
      ],  
      [  
        4,  
        "2125-07-01 00:00:00"  
      ]  
    ],  
    [],  
    [  
      0,  
      "",  
      "",  
      0,  
      ""  
    ]  
  ]  
]
```

The endpoint-specific information includes:

```
unknown = 0
unknown = ""
unknown = [
  [
    1,
    "2100-01-01 07:00:00"
  ],
  [
    2,
    "2100-01-01 07:15:00"
  ],
  [
    3,
    "2100-01-01 07:25:00"
  ],
  [
    4,
    "2125-07-01 00:00:00"
  ]
]
unknown = []
unknown = [
  0,
  "",
  "",
  0,
  ""
]
```

Warning

All the values are currently of unknown purpose. Further research is required to determine its intended use.

4.3.2 adjustment_data_manage/read

4.3.2.1 Request

```
POST /025348/api/adjustment_data_manage/read HTTP/2
:authority: dbtb-prd.cosmos.channel.or.jp
content-type: application/x-www-form-urlencoded
content-length: 90
```

The request contained the common user fields and no additional fields.

```
[
  {
    "titleCd": "025348",
    "userId": "685129260522192549",
    "session": "6a10adbf368d5",
    "platform": 3,
    "version": "09.01"
  },
  []
]
```

4.3.2.2 Response

The response headers are:

```
HTTP/2 200 OK

content-type: application/x-messagepack; charset=utf-8
content-length: 56411
```

The response contained the common user fields with additional fields.

```
[
  {
    "result": 0,
    "date": "2026/05/22 19:26:09",
    "version": "09.01",
    "flag": "0",
    "session": "6a10add143ef7"
  },
  [
    0,
    2,
    "T0JE0i6kl9BkRCHhvs ... <More data>"
  ]
]
```

The endpoint-specific information includes:

```
unknown  = 0
unknown  = 2
payload  = "T0JE0i6kl9BkRCHhvs ... <More data>"
```

Warning

All the values are currently of unknown purpose. Further research is required to determine its intended use.

4.3.2.3 Payload

The payload is composed almost entirely of printable ASCII characters.

Examples from the capture include:

```
T0JE0i6k19BkRCHvs-SEnG9ivUPaBK8bGTdTszz_-H86_aBA4a...
```

and highly repetitive sequences such as:

```
02ja96M7aNr3ozto2vej02ja96M7aNr3ozto2vej...
```

The observed character set includes:

```
A-Z  
a-z  
0-9  
-  
-
```

This resembles an ASCII-safe or Base64URL-like representation.

However, the packet capture alone does not establish that the data is standard Base64URL.

It may instead represent:

1. compressed application data encoded into ASCII,
2. a custom game serialization,
3. a custom binary-to-text representation,
4. encrypted or obfuscated data,
5. or a combination of the above.

Therefore, this paper treats the field as an *opaque 56,327-byte encoded data blob*.

4.4 Friends and Social

Endpoint	Function description
friend/get_friend_list	Friend-list retrieval
friend/send_friend_request	Friend request submission
friend/accept_friend_request	Friend request acceptance
friend/delete_friend	Friend removal
friend/get_platform_friend	Platform friend lookup
friend/update_favorite_status	Favorite-status update
game_setting/update_user_setting	User game-setting update

4.5 Blocking

Endpoint	Function description
<code>block/get_block_list</code>	Block-list retrieval
<code>block/add_block_user</code>	User blocking
<code>block/delete_block_user</code>	Block removal

4.6 Battle

Endpoint	Function description
battle/get_diarkis_matching_server_info	Diarkis matching-server information
battle/get_connection_server_info	Connection-server information
battle/get_stun_server_info	STUN-server information
battle/save_matching_cache	Matching-cache storage
battle/start	Battle start
battle/result	Battle result submission
battle/practice_start	Practice battle start
battle/practice_result	Practice battle result submission
battle/pre_matching_connection	Pre-matching connection
battle/get_battle_member_result_list	Battle-member result retrieval
battle/consume_priority_point	Priority-point consumption
battle/waittime_preview	Battle wait-time preview
battle/orientation_start	Orientation battle start
battle/orientation_result	Orientation battle result submission

4.6.1 battle/get_stun_server_info

4.6.1.1 Request

```
POST /025348/api/battle/get_stun_server_info HTTP/2
:authority: dbtb-prd.cosmos.channel.or.jp
content-type: application/x-www-form-urlencoded
content-length: 90
```

The request contained the common user fields without any additional fields.

```
[
  {
    "titleCd": "025348",
    "userId": "685129260522192549",
    "session": "6a10add4e2cda",
    "platform": 3,
    "version": "09.01"
  },
  []
]
```

4.6.1.2 Response

The response headers are:

```
HTTP/2 200 OK

content-type: application/x-messagepack; charset=utf-8
content-length: 56411
```

The response contained the common user fields with additional fields.

```
[
  {
    "result": 0,
    "date": "2026/05/22 19:26:31",
    "version": "09.01",
    "flag": "0",
    "session": "6a10ade71a276"
  },
  [
    0,
    [
      [
        "dbform-prd-025348-stun01.cosmos.channel.or.jp",
        3478
      ],
      [
        "dbform-prd-025348-stun02.cosmos.channel.or.jp",
        3478
      ]
    ]
  ]
]
```

The endpoint-specific information includes:

```
unknown = 0
stun_servers = [
  "dbform-prd-025348-stun01.cosmos.channel.or.jp",
  3478
],
[
  "dbform-prd-025348-stun02.cosmos.channel.or.jp",
  3478
]
]
```

Warning

The first value is currently of unknown purpose. Further research is required to determine its intended use.

Info

<https://en.wikipedia.org/wiki/STUN>

4.7 Events

Endpoint	Function description
event/get_schedule_list	Scheduled-event list retrieval
event_tier/get_event_tier_info	Event-tier information retrieval
event_tier/earn_tier_reward	Event-tier reward acquisition

4.8 Friend Codes

Endpoint	Function description
friend_code/create_friend_code	Friend-code creation
friend_code/enter_friend_code	Friend-code entry

4.9 Lootbox

Endpoint	Function description
lootbox/get_transball_list	Transball lootbox list
lootbox/get_transball_probability	Transball probability information
lootbox/run_transball	Execute Transball lootbox
lootbox/user_ticket_list	User ticket list
lootbox/ticket_master_list	Ticket master-data list

4.10 Leaderboard

Endpoint	Function description
leaderboard/get_leaderboard_model	Leaderboard 3D-model information
leaderboard/get_leaderboard_group_list	Leaderboard-group list
leaderboard/get_leaderboard	Leaderboard data

4.11 Messages

Endpoint	Function description
message/get_message_list	Message-list retrieval
message/get_message_info	Message-detail retrieval
message/update_message_item_received	Message-item receipt update

4.12 News

Endpoint	Function description
news/get_3d_model_news	3D news-model retrieval
news/get_2d_ui_news_list	2D news/UI list retrieval

4.13 Character and Cosmetic Data

Endpoint	Function description
character/get	Character data retrieval
character/get_item	Character item retrieval
noncombat_chara_skin/get_list	Non-combat character-skin list
costume/get_list	Costume list
emotion/get_list	Emotion list
stamp/get_list	Stamp list
intro_demo/get_list	Introduction-demo list
victorypose/get_list	Victory-pose list
vehicleskin/get_list	Vehicle-skin list
avatar_create/get_list	Avatar creation-part list
user/update_avatar	Avatar thumbnail/image update

4.14 Transball and Skills

Endpoint	Function description
transball/get_list	Transball list
transball/unlock_spattack	Special-attack unlock
transball/get_all_chara	Transball character release information
skill/get_skill_list	Skill list
skill/training	Skill training

4.15 Rival and Patroller Systems

Endpoint	Function description
rival/get_list	Rival list retrieval
rival/training	Rival skill training
rival/skill_reset	Rival skill reset
patroller/get_status	Patroller status retrieval

4.16 Player and Economy

Endpoint	Function description
player/upload_ghost_player	Ghost-player upload
gamecurrency/get_owned	Owned game-currency retrieval
item/item_possession	Item-possession information
episode/earn_episode_reward	Episode reward acquisition
serial_code/get_serial_code_item	Serial-code item retrieval

4.17 Season

Endpoint	Function description
season/get_selectable_stage	Selectable battle-stage retrieval
season/get_dragon_pass_info	Dragon Pass information
season/tier_up_using_tp	Tier purchase using TP
season/earn_dragon_pass_reward	Dragon Pass reward acquisition

4.18 Shop

Endpoint	Function description
shop/get_category_list	Shop-category list
shop/get_info_category	Shop-category information
shop/get_info_special_category	Special shop-category information
shop/payment	Shop-item payment/purchase

4.19 Common Purchase

Endpoint	Function description
<code>commonpurchase/tokusho</code>	Special commercial-information display
<code>commonpurchase/get_fund_settlement_text</code>	Fund-settlement text retrieval
<code>commonpurchase/get_catalog</code>	Purchase catalog retrieval
<code>commonpurchase/start_purchase</code>	Purchase initiation
<code>commonpurchase/complete_purchase</code>	Purchase completion
<code>commonpurchase/get_purchase_status</code>	Purchase-status retrieval
<code>commonpurchase/cancel_purchase</code>	Purchase cancellation

4.20 Online Battle and Connection Data

Endpoint	Function description
battle/get_diarkis_matching_server_info	Diarkis matching-server information
battle/get_connection_server_info	Connection-server information
battle/get_stun_server_info	STUN-server information
battle/save_matching_cache	Matching-cache storage
adjustment_data_manage/get_version	Online battle data version
adjustment_data_manage/read	Online battle data retrieval

4.21 Miscellaneous Battle Configuration

Endpoint	Function description
battle/get_challenge_list	Challenge list
battle/get_battle_member_result_list	Battle-member result list
battle/consume_priority_point	Priority-point consumption
battle/waittime_preview	Wait-time preview
character/get	Character retrieval

5 End-to-End Request Sequence

Combining the observed calls gives the following approximate application sequence:

```
1. /000000/api/user/auth
    |
    | authentication
    v
    session established
    |
    v
2. /000000/api/user/get_country
    |
    | country lookup
    v
    "GB"
    |
    v
3. /000000/api/user/get_tracking_num
    |
    | tracking number retrieval
    v
    "LXDA-45J98PDC7NL"
    |
    v
4. /025348/api/close/get_close_info
    |
    | language = "en"
    v
    close/status configuration
    |
    v
5. /025348/api/user/create_user_info
    |
    | country = "GB"
    | language = "en"
    | Steam ID
    v
    user information processing
    |
    v
6. /025348/api/adjustment_data_manage/read
    |
    | no endpoint arguments
    v
    56,327-byte adjustment-data blob
```

The exact ordering shown here follows the sequence of captures supplied for analysis; it should not be interpreted as a complete application startup state machine.

6 Conclusions

The packet captures establish a coherent HTTP/2 MessagePack application protocol.

The most significant properties are:

1. All observed API operations use HTTP/2 POST.
2. The APIs are transported over HTTPS.
3. MessagePack is used for the application data.
4. Requests use a recurring client context containing title, user, session, and platform information.
5. Game-specific requests additionally contain a version field.
6. Endpoint-specific arguments are represented as a second array.
7. Successful responses use HTTP status 200.
8. Responses contain a common result/date/session metadata object.
9. The session value evolves across successive requests.
10. The country API returns GB.
11. The tracking-number API returns LXDA-45J98PDC7NL.
12. The user-information API supplies country, language, and Steam ID.
13. The close-information API accepts a language argument of "en".
14. The adjustment-data API returns an opaque 56,327-byte encoded string inside a 56,411-byte MessagePack response.

The resulting high-level protocol can therefore be summarized as:

```
Client
|
| POST /endpoint
| common MessagePack context
| endpoint arguments
v
Server
|
| 200 OK
| result metadata
| updated session
| endpoint-specific data
v
Client
```

7 Reverse engineering

7.1 Executable

7.1.1 API

7.1.1.1 Class Architecture

Reverse engineering of the executable reveals a systematic family of request and response classes.

A typical request entry has the following conceptual structure:

```
.rdata
  RTTI Complete Object Locator
  vftable
    virtual function
    virtual function
    ...
  "ServerApiRequest<Operation>"
  "<api endpoint>"
```

For example, the executable contains:

```
ServerApiRequestGetEnvironmentV3
|
+-- "ServerApiRequestGetEnvironmentV3"
|
+-- "sys/get_env_v3"
```

and the corresponding response:

```
ServerApiResponseGetEnvironmentV3
|
+-- "ServerApiResponseGetEnvironmentV3"
```

This provides a useful relationship:

$$\text{Request Class} \longleftrightarrow \text{API Endpoint} \longleftrightarrow \text{Response Class} \quad (1)$$

The class names and route strings are consequently useful for reconstructing the logical API surface even when the detailed field layouts have not yet been recovered.

The captures provide enough information to reconstruct the HTTP/2 and MessagePack framing for the observed calls. The remaining unknowns are primarily the semantic definitions of some endpoint-specific fields and the internal format of the large adjustment-data blob.

The following table lists the request classes, endpoint strings, and matching response classes recovered from the executable material.

7.1.1.2 Class list

#	Operation	API Endpoint
1	GetAvatarCreatePartsList	avatar_create/get_list
2	GetBattleMemberResult	battle/get_battle_member_result_list
3	GetChallengeList	battle/get_challenge_list
4	GetConnectionServerInfo	battle/get_connection_server_info
5	GetDiarkisMatchingServerInfo	battle/get_diarkis_matching_server_info
6	GetStunServerInfo	battle/get_stun_server_info
7	BattleResult	battle/result
8	SaveMatchingCache	battle/save_matching_cache
9	BattleStart	battle/start
10	ConsumePriorityPoint	battle/consume_priority_point
11	OrientationBattleResult	battle/orientation_result
12	OrientationBattleStart	battle/orientation_start
13	PreMatchingConnection	battle/pre_matching_connection
14	PracticeBattleResult	battle/practice_result
15	PracticeBattleStart	battle/practice_start
16	AddBlockUser	block/add_block_user
17	DeleteBlockUser	block/delete_block_user
18	GetBlockList	block/get_block_list
19	GetCloseInfo	close/get_close_info
20	GetFundSettlement	commonpurchase/get_fund_settlement_text
21	GetPurchaseCatalog	commonpurchase/get_catalog
22	GetPurchaseStatus	commonpurchase/get_purchase_status
23	CompletePurchase	commonpurchase/complete_purchase
24	CancelPurchase	commonpurchase/cancel_purchase
25	StartPurchase	commonpurchase/start_purchase
26	ShowFundSettlement	commonpurchase/get_fund_settlement_text
27	ShowTokusho	commonpurchase/tokusho
28	GetCostumeList	costume/get_list
29	ShowBattleSettings	character/get
30	ShowCharacterVisual	character/get_item
31	GetEventDragonTierInfo	event_tier/get_event_tier_info
32	ReceiveEventTierReward	event_tier/earn_tier_reward
33	CreateFriendCode	friend_code/create_friend_code
34	InputFriendCode	friend_code/enter_friend_code
35	AcceptFriendOrder	friend/accept_friend_request
36	DeleteFriend	friend/delete_friend
37	GetFriendList	friend/get_friend_list
38	GetPlatformFriendList	friend/get_platform_friend
39	ChangeFriendStatus	friend/update_favorite_status
40	SendFriendOrder	friend/send_friend_request
41	GetTpToken	gamecurrency/get_owned
42	UpdateSocialOption	game_setting/update_user_setting
43	GetIntroDemoList	intro_demo/get_list
44	PossessionPurchaseItem	item/item_possession
45	GetGashaList	lootbox/get_transball_list
46	GetGashaLineup	lootbox/get_transball_probability
47	ExecuteGasha	lootbox/run_transball
48	GetGashaTicketMasterList	lootbox/ticket_master_list
49	GetGashaTicketList	lootbox/user_ticket_list
50	GetMessageDetail	message/get_message_info
51	GetMessageList	message/get_message_list
52	ReceiveMessageItem	message/update_message_item_received
53	GetNewsBoardTexture	news/get_3d_model_news
54	GetNewsTextureList	news/get_2d_ui_news_list

#	Operation	API Endpoint
55	GetNonCombatCharacterList	noncombat_chara_skin/get_list
56	RegistPlayedWithPlayer	player/upload_ghost_player
57	GetPatrollerSkillList	skill/get_skill_list
58	ExecuteSkillTraining	skill/training
59	GetSelectableBattleStage	season/get_selectable_stage
60	GetDragonPassInfo	season/get_dragon_pass_info
61	BuyTier	season/tier_up_using_tp
62	ReceiveTierReward	season/earn_dragon_pass_reward
63	GetShopCategoryList	shop/get_category_list
64	GetShopCategoryDetail	shop/get_info_category
65	GetShopSpecialCategoryDetail	shop/get_info_special_category
66	BuyShopItem	shop/payment
67	InputSerialCode	serial_code/get_serial_code_item
68	AgreeKPI	sys/agree_kpi
69	CheckNgName	sys/check_ngname
70	GetEnvironmentV3	sys/get_env_v3
71	KPI	sys/kpi
72	ExecuteUnlockSpAttack	transball/unlock_spattack
73	GetTransballReleaseDateList	transball/get_all_chara
74	GetTransballList	transball/get_list
75	GetBannStatus	user/get_ban_status
76	GetCountryCode	user/get_country
77	CreateUser	user/create_user_info
78	GetUserTrackingNumber	user/get_tracking_num
79	UpdateAvatarThumbnailImage	user/update_avatar
80	UpdateMannerPoint	user/update_manner_point
81	UpdateOrientationStep	user/update_orientation_step
82	UserAuth	user/auth
83	GetVehicleSkinList	vehicleskin/get_list
84	GetVictoryPoseList	victorypose/get_list
85	GetScheduleEventList	event/get_schedule_list
86	GetEpisodeReward	episode/earn_episode_reward
87	GetRivalList	rival/get_list
88	ExecuteRivalSkillReset	rival/skill_reset
89	ExecuteRivalSkillTraining	rival/training
90	CheckBnidLinkStatus	bnid_client/check_status
91	GetBnidLinkURL	bnid_client/entry
92	EraseBnidLink	bnid_client/erase
93	GetBnidLinkReward	bnid_reward/grant_reward
94	GetOnlineBattleData	adjustment_data_manage/read
95	GetOnlineBattleDataVersion	adjustment_data_manage/get_version
96	CheckMaintenance	regularly_run/regularly_maintenance_check
97	GetEmergencyMessage	regularly_run/get_emergency_message
98	UnfairPlayReport	report/report
99	GetEpisodeReward	episode/earn_episode_reward
100	GetStampList	stamp/get_list
101	GetEmotionList	emotion/get_list
102	GetTpToken	gamecurrency/get_owned
103	GetLeaderBoard	leaderboard/get_leaderboard
104	GetLeaderBoardGroup	leaderboard/get_leaderboard_group_list